SATELLITE ADCS TELEMETRY OPERATIONS MANUAL

ESA OPS-SAT CubeSat Mission ? Attitude Determination and Control System
Version 1.0 | Ground Operations Reference

1. INTRODUCTION AND SCOPE

This manual serves as the primary operations reference for monitoring, diagnosing, and responding to anomalies detected in the Attitude Determination and Control System (ADCS) telemetry of the ESA OPS-SAT CubeSat mission. It covers the nine primary ADCS telemetry channels: CADC0872, CADC0873, CADC0874 (magnetometers X, Y, Z) and CADC0875, CADC0876, CADC0877, CADC0878, CADC0879, CADC0880 (photodiodes 1-6).

The OPS-SAT satellite is a 3-Unit CubeSat (10 x 10 x 30 cm, ~5.4 kg) launched in December 2019. Its ADCS combines a coarse attitude system on the bus and a fine ADCS on the payload, including star tracker, reaction wheels, magnetorquers, gyroscopes, magnetometers, and photodiodes.

2. ADCS SUBSYSTEM ARCHITECTURE

2.1 Sensors

MAGNETOMETERS (CADC0872, CADC0873, CADC0874):
- Three-axis fluxgate/Hall-effect sensors measuring Earth's magnetic field vector in the satellite body frame
- Provide attitude reference by comparing measured field with IGRF model
- Critical for magnetorquer control: without good magnetometer data, magnetic torquing is impossible
- Nominal measurement range: ±60 ?T (microTesla) in LEO
- Nominal sampling: every telemetry cycle
- Channels: X-axis (CADC0872), Y-axis (CADC0873), Z-axis (CADC0874)

PHOTODIODES (CADC0875 ? CADC0880):
- Six coarse sun sensors covering ±X, ±Y, ±Z faces of the satellite
- Each photodiode measures solar irradiance incident on its face
- Used for sun vector determination and solar panel orientation
- In eclipse: all readings drop to near zero ? this is NOMINAL behaviour
- Outputs are analog voltage proportional to light intensity
- Channels: +X face (CADC0875), -X face (CADC0876), +Y face (CADC0877), -Y face (CADC0878), +Z face (CADC0879), -Z face (CADC0880)

2.2 Actuators

MAGNETORQUERS (3-axis):
- Generate magnetic dipole moment to interact with Earth's field
- Used for de-tumbling, fine attitude control, reaction wheel desaturation
- Emit EMI that temporarily corrupts magnetometer readings during operation
- Nominal operating current: 50?200 mA per axis

REACTION WHEELS (3-axis):
- Store and exchange angular momentum with the spacecraft body
- Speed range: typically ±6000 RPM
- Saturation occurs when wheel reaches maximum speed and can no longer provide additional torque
- Require periodic desaturation via magnetorquers

STAR TRACKER:
  - High-precision attitude sensor matching star patterns to a catalog
  - Requires unobstructed field of view and sufficient star count
  - Lost lock causes degraded attitude determination

2.3 ADCS Operating Modes

DETUMBLE MODE:
  - Activated after launch or safe mode entry
  - Uses B-dot controller with magnetorquers to reduce tumbling rates
  - Magnetometer readings are critical during this phase
  - Photodiode readings may be highly variable due to tumbling

SUN-POINTING MODE:
  - Aligns a chosen body axis toward the Sun
  - Relies primarily on photodiode sun vector measurements
  - Used for power generation and thermal management

NADIR-POINTING MODE:
  - Aligns a chosen body axis toward Earth centre
  - Uses star tracker and reaction wheels for precision pointing
  - Magnetometers used for reference and magnetorquer desaturation

SAFE MODE:
  - Minimum-power configuration entered on fault detection
  - Only coarse attitude control active
  - All non-essential payloads powered down
  - Sun-pointing to maintain power generation

3. TELEMETRY CHANNEL NOMINAL RANGES
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Channel	Sensor	Nominal Range	Alert Threshold
CADC0872	Magnetometer X	-60 to +60 ?T	±65 ?T
CADC0873	Magnetometer Y	-60 to +60 ?T	±65 ?T
CADC0874	Magnetometer Z	-60 to +60 ?T	±65 ?T
CADC0875	Photodiode +X	0 to 1.2 V (sunlit)	>1.4 V
CADC0876	Photodiode -X	0 to 1.2 V (sunlit)	>1.4 V
CADC0877	Photodiode +Y	0 to 1.2 V (sunlit)	>1.4 V
CADC0878	Photodiode -Y	0 to 1.2 V (sunlit)	>1.4 V
CADC0879	Photodiode +Z	0 to 1.2 V (sunlit)	>1.4 V
CADC0880	Photodiode -Z	0 to 1.2 V (sunlit)	>1.4 V

Note: All photodiode channels reading zero simultaneously during expected
sunlit pass is an anomaly, not normal eclipse behaviour.

4. OPERATIONAL TIMELINE AND CONTEXT
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OPS-SAT orbits at approximately 520 km altitude, inclination ~97°, in a
Sun-synchronous orbit. Orbital period is approximately 95 minutes.
Eclipse duration: ~35 minutes per orbit.

Telemetry downlink: S-band, 1 Mbit/s downlink during ground station passes.
Ground contact windows: approximately 3-4 passes per day, 8-12 minutes each.

Between passes, telemetry is stored on board and bulk-downlinked during
the next contact window. Anomalies detected in stored telemetry may have
occurred up to 90 minutes before the operator becomes aware of them.

This time lag is a critical operational constraint: corrective commands must account for the fact that the anomaly state may have evolved significantly before any uplink can be sent.

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END OF SECTION 1 ? ADCS OPERATIONS MANUAL

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